# Ahmed Ezzat


DevOps Engineer

+20 120 411 4788 | ahmedezzatzezo7@gmail.com | linkedin.com/in/ahmed-ezzat-a6b84b277 | github.com/ahmedezzat-7


## Summary


AI & Data Engineer with hands-on experience designing distributed data platforms and building scalable batch and real-time pipelines using Python, Apache Kafka, and Apache Spark. Experienced in implementing Change Data Capture (CDC) pipelines with Debezium, orchestrating ETL/ELT workflows using Apache Airflow and dbt, and developing modern lakehouse architectures Skilled in AWS data services, Snowflake, BigQuery, and Redshift to deliver reliable data platforms for analytics and machine learning and AI workloads


## Education

**Mansoura University** Sep 2019 – Jul 2023
*BSc. in Computer and Information Science*

## Experience

**Freelance Data & AI Engineer - 2 months** 02/2026 – 04/2026
*Social Media Marketing Intelligence Platform* *Remote*

- Ingested live social media data from the Meta API using an Apache Kafka cluster (3 brokers + Schema Registry) to ensure fault-tolerant, schema-governed event streaming
- Processed streaming data with Spark Structured Streaming (1 master, 3 workers) across a Medallion Architecture (Bronze , Silver , Gold), enforcing data quality
- Persisted Gold-layer data as Apache Iceberg tables on Azure, enabling ACID transactions and time-travel queries
- served via Redis cache and Streamlit dashboard
- Containerized the full pipeline using Docker and deployed the entire infrastructure on Azure cloud

## Projects

**Banking Real-Time Streaming Data Platform** Link
*Python, PostgreSQL, Debezium, Kafka, Airflow, Snowflake, dbt, Docker, GitHub Actions*

- Engineered a CDC pipeline with Debezium and Kafka to capture real-time database changes from PostgreSQL
- Orchestrated 4+ pipelines in Airflow to load data into Snowflake, using dbt for SCD Type 2 modeling
- Automated 7-service containerized infrastructure via Docker and GitHub Actions CI/CD for deployment

**Smart City Real-Time Streaming Platform** Link
*Python, Kafka, Spark, AWS (S3, Glue, Athena, Redshift), Docker, Terraform, Power BI*

- Ingested 5 IoT data streams using Kafka and Spark Structured Streaming for real-time urban mobility analysis
- Built an AWS-based Data Lake (S3, Glue, Athena) with Redshift for large-scale analytics and testing
- Managed cloud infrastructure with Terraform and monitored system health via Grafana and Power BI dashboards

## Technical Skills

**Programming & Libraries:** Python (Pandas, NumPy, Matplotlib, Seaborn, SciPy), SQL

**AI & Machine Learning:** Scikit-learn, PyTorch, XGBoost, Supervised/Unsupervised Learning, Deep Learning, NLP,

**Big Data & Streaming:** Apache Spark (PySpark, Structured Streaming,MLlib), Apache Kafka, Apache Flink , Apache Hadoop,HDFS

**Orchestration & ETL:** Apache Airflow, dbt, ETL/ELT Pipelines, Change Data Capture (CDC), Debezium

**Cloud (AWS & Azure):** S3, Glue, Athena, Redshift, Lambda, RDS, EC2, EMR, Azure Blob Storage

**Data Warehousing & Lakehouse:** Snowflake, BigQuery, Apache Iceberg, Medallion Architecture, Lakehouse Architecture

**Databases :** PostgreSQL, SQL Server, NoSQL

**Data Analysis & Visualization:** Power BI, Grafana , EDA, Data Preprocessing

**DevOps & Infrastructure:** Linux, Docker, Kubernetes, Git, CI/CD (GitHub Actions), Terraform

**Data Modeling:** Dimensional Modeling, Star Schema, Snowflake Schema, OLAP/OLTP, SCD Type 2

**Core Concepts:** Data Pipelines, Distributed Systems, Stream Processing, Batch Processing, Data Platform Architecture, Feature Engineering, Model Evaluation

## Languages

**Arabic:** Native — **English:** Professional Working Proficiency